

Practical 3

Practical 3: MongoDB Operations

Aim:

To perform single and multiple document insertion and retrieval using the MongoDB shell (mongosh).

Code:

```
// How to run: mongosh
//          > use practical3
//          > load("practical_03_mongodb.js")

// >>> CHANGE THIS TO YOUR NAME <<<
const STUDENT_NAME = "Purvi Prakash Piprikar";

// Start clean so re-running does not create duplicate documents
db.items.drop();
db.products.drop();

// -----
print("\n## Step 1: Insert Single Document");
print("Query: db.items.insertOne({ name: \"laptop\", price: 999 });");
printjson(db.items.insertOne({ name: "laptop", price: 999 }));

// -----
print("\n## Step 2: View Inserted Document");
print("Query: db.items.find();");
printjson(db.items.find().toArray());

// -----
print("\n## Step 3: Insert Multiple Documents");
```

```
print("Query: db.products.insertMany([...3 products...]);");
printjson(db.products.insertMany([
    { name: "phone", price: 500, stock: 10 },
    { name: "tablet", price: 300, stock: 5 },
    { name: "watch", price: 150, stock: 0 }
]));
```

```
// -----

print("\n## Step 4: View Products in Formatted JSON");
print("Query: db.products.find().pretty();");
printjson(db.products.find().toArray());
```

```
// -----

print("\n=====");
print("Submitted by: " + STUDENT_NAME);
print("=====");
```

Output:

I/O AI Agent 1.0s ✓

STDIN

Input for the program (Optional)

Output:

true

true

Step 1: Insert Single Document

Query: db.items.insertOne({ name: "laptop", price: 999 });

{

acknowledged: true,

insertedId: ObjectId('6ac9fad8dfdb0509e0e91cdc')

}

Step 2: View Inserted Document

Query: db.items.find();

```
[
  {
    _id: ObjectId('6ac9fad8dfdb0509e0e91cdc'),
    name: 'laptop',
    price: 999
  }
]
```

Step 3: Insert Multiple Documents

Query: db.products.insertMany([...3 products...]);

```
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6ac9fad8dfdb0509e0e91cdd'),
    '1': ObjectId('6ac9fad8dfdb0509e0e91cde'),
    '2': ObjectId('6ac9fad8dfdb0509e0e91cdf')
  }
}
```

Step 4: View Products in Formatted JSON

Query: db.products.find().pretty();

```
[
  {
    _id: ObjectId('6ac9fad8dfdb0509e0e91cdd'),
    name: 'phone',
    price: 500,
    stock: 10
  },
  {
    _id: ObjectId('6ac9fad8dfdb0509e0e91cde'),
    name: 'tablet',
    price: 300,
    stock: 5
  },
  {
    id: ObjectId('6ac9fad8dfdb0509e0e91cdf'),
```

Output:

```
  },  
  {  
    _id: ObjectId('6ac9fad8dfdb0509e0e91cde'),  
    name: 'tablet',  
    price: 300,  
    stock: 5  
  },  
  {  
    _id: ObjectId('6ac9fad8dfdb0509e0e91cdf'),  
    name: 'watch',  
    price: 150,  
    stock: 0  
  }  
]
```

Submitted by: Purvi Prakash Piprikar